

Camila VILLALBA HEUER

Java Backend Developer (Junior) |
SQL & Relational Databases



PROFESSIONAL SUMMARY

Java Backend Developer (Junior) with an engineering background and hands-on experience gained through academic and self-directed projects. Focused on building backend systems with Java, Spring Boot, SQL, and microservices architecture, applying clear validation rules, explicit state management, and well-documented REST APIs, with experience implementing Spring Security and JWT-based authentication, along with backend testing practices. Background in safety-critical environments (nuclear engineering), bringing a systems-thinking mindset oriented to reliability, data consistency, and controlled processes. Motivated to continue growing as a backend developer through real-world projects and collaborative teams.

CONTACT

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PERSONAL INFORMATION

Age: **30 years old**

Nationality: **Argentine**

LANGUAGES

Spanish (Native)

English (B1)

SOFTWARE PROJECTS

January 2026

🔗 Microservices Architecture - Online Appliance Store (Spring Boot + Spring Cloud + Docker)

Designed and implemented a distributed e-commerce system based on microservices architecture, applying Spring Cloud patterns to ensure scalability, fault tolerance, and data consistency.

- ◇ Built a microservices ecosystem with Eureka (Service Discovery), API Gateway, Config Server, and OpenFeign for inter-service communication.
- ◇ Implemented fault tolerance with Resilience4j and enforced explicit state management for carts (CREATED, SOLD) and products (ACTIVE, DISCONTINUED) to prevent inconsistent operations.
- ◇ Applied Bean Validation on DTOs, global exception handling, and documented REST APIs with Swagger (OpenAPI).
- ◇ Implemented distributed security with Spring Security and JWT, separating user authentication/authorization from service-to-service security through a dedicated technical service identity and role-based access control.
- ◇ Applied controlled state transitions inspired by safety-critical engineering systems, ensuring consistency without tight inter-service coupling.
- ◇ Refactored DTOs to Java records and separated request/response models to improve code clarity and maintainability.

October 2025

🔗 Java Desktop Application – Grooming Manager (Swing + JPA + MySQL)

Designed and developed a Java-based desktop management system applying layered architecture principles and strong data consistency rules.

- ◇ Built a Java Swing application with MVC-style layered architecture (UI, business logic, persistence).
- ◇ Modeled domain entities with complex relationships and implemented advanced data validation to prevent duplicated or inconsistent records.
- ◇ Designed persistence layer using JPA (EclipseLink) with MySQL, managing ORM mappings and transactional operations.
- ◇ Managed the project with Maven and implemented usability improvements such as autocomplete for data entry.

TECHNICAL SKILLS

- ◇ Backend Development
 - Java SE (JDK 17): OOP, Collections, Functional Programming (Streams, Lambda expressions, Method References), Exception Handling, JDBC
 - Java EE / Jakarta EE concepts: Servlets, JSP, JSTL, JPA
 - Spring Boot: REST APIs development, MVC architecture, validation, exception handling
 - Spring Data JPA / Hibernate / EclipseLink
 - Spring Cloud: Service Discovery (Eureka), API Gateway, Config Server, client-side load balancing
 - Microservices architecture (academic projects)

ADDITIONAL PROJECTS

🔗 REST API - Sales & Inventory Management System

Spring Boot · JPA · MySQL / H2

- Designed and implemented a RESTful API applying MVC architecture, DTO pattern, and business validations.
- Developed full CRUD operations with relational persistence and transactional consistency.
- Applied layered architecture and clean separation between controllers, services, and repositories.

🔗 Supermarket Management System

Java · functional programming

- Implemented a backend-focused Java application applying functional programming concepts (streams, lambdas, method references)
- Developed business logic for data processing, filtering, aggregation, and transformation.

🔗 Python Mini-Projects Collection

Python · Testing · CI

- Developed multiple backend-oriented Python applications, including an inventory management system with dual persistence (in-memory / MySQL).
- Implemented automated testing (90+ tests) and CI pipelines using GitHub Actions.
- Applied modular design principles to support scalability and maintainability.

🔗 Relational Database Design Projects

MySQL · SQL

- Designed and implemented relational database schemas for e-commerce, bookshop, and clinic systems.
- Wrote advanced SQL queries using joins, subqueries, aggregations, and transactions.
- Implemented stored procedures and indexes to encapsulate business logic and improve performance.

SOFT SKILLS

- Analytical thinking
- Problem-solving
- Attention to detail
- Fast learning & adaptability
- Clear communication
- Methodical & organized
- Team collaboration

◇ Security & Authentication

- Spring Security with JWT-based authentication and authorization
- Role and permission-based access control
- OAuth2 fundamentals
- Auth0 integration (in progress)
- JWT library for token generation and validation

◇ Databases and Persistence

- SQL (MySQL): relational modeling, joins, subqueries, aggregations, transactions
- Constraints, foreign keys, indexes
- Stored procedures and basic triggers
- ORM mapping and transactional management with JPA

◇ API & Integration

- RESTful API design and integration
- DTO pattern for request/responses handling
- API documentation: Swagger / OpenAPI
- API testing: Postman (Collections)
- Inter-service communication using OpenFeign

◇ Configuration & Environments

- Application configuration using YAML
- Environment-specific profiles (local / Docker)

◇ Build, DevOps & Tools

- Maven (multi-module projects, dependency management)
- Docker (Dockerfile creation, containerization basics)
- Git / GitHub (version control, repositories)
- Apache Tomcat

◇ Frontend Basics

- HTML5, CSS3, JavaScript
- Java Swing (desktop applications)

◇ Others Programming

- Python: Pandas, Numpy, Django

◇ Agile & Development Practices

- Scrum methodology: understanding of roles, ceremonies, artifacts, and iterative development

ADDITIONAL TRAINING

◇ Backend Development (TodoCode Academy): Microservices with Spring Cloud, API Development with Spring Boot, Java Web (JSP, Servlets, JPA), Functional Programming, Object-Oriented Programming, Software Testing and Quality, Security in Java with Spring Security

◇ Databases: SQL (MySQL) – Self-taught (SQLBolt, DB Fiddle, XAMPP + phpMyAdmin)

◇ Python: Introduction to Python, Python for Data Analysis (Pandas, NumPy), Testing & CI/CD with Python (unittest, GitHub Actions)

◇ Web Basics: HTML, CSS, JavaScript

◇ Data & Analytics: Power BI and Power Query (Santander Academy), Introduction to Data Analysis (Unicorn Academy)

PROFESSIONAL MOTIVATION

Motivated by designing reliable backend systems and clean architectures using Java and Spring Boot. I enjoy modeling real-world domains, enforcing data consistency, and implementing validation and transactional logic that ensures system integrity.

Driven by continuous learning and the application of engineering principles to software development, with the goal of growing as a Java Backend Developer and contributing to well-structured, maintainable systems.

RELEVANT INTERESTS

Interested in backend architecture, REST API design, and the Java/Spring Boot ecosystem. Particularly motivated by relational database design, SQL optimization, ORM usage (JPA/Hibernate), and improving system reliability through explicit state management and clean data modeling. Interested in practices that enhance code quality and maintainability, such as clean code principles, testing, and version control workflows.



HEAD OF MAINTENANCE at RA-10 Nuclear Reactor – CNEA

- ◇ Designed and maintained structured technical documentation and workflows for maintenance processes in a safety-critical environment.
- ◇ Analyzed equipment data using structured datasets (tables, indicators, classifications) to support decision-making and operational planning.
- ◇ Defined consistency rules and validation criteria for maintenance information, ensuring reliable and auditable records.
- ◇ Developed dashboards and KPIs to track activities, detect deviations, and support corrective actions.
- ◇ Collaborated with interdisciplinary technical teams, applying analytical and methodical problem-solving approaches.
- ◇ Applied systems-thinking principles from nuclear safety engineering, focusing on controlled processes, traceability, and error prevention.

March 2023 – June 2024



HEAD OF OPERATIONS at RA-10 Nuclear Reactor – CNEA

- ◇ Validated operational and maintenance manuals to ensure internal consistency and compliance with licensing requirements.
- ◇ Defined critical operational variables and designed structured data collection spreadsheets for monitoring and analysis.
- ◇ Standardized information flows to reduce ambiguity and improve data reliability across operational processes.
- ◇ Supported decision-making through structured analysis of operational data and documented procedures.

EDUCATION

February 2022 – December 2022



SPECIALIZATION IN TECHNOLOGICAL APPLICATIONS OF NUCLEAR

ENERGY at *Balseiro Institute and Faculty of Engineering – University of Buenos Aires (UBA)*

February 2015 – September 2021



CHEMICAL ENGINEERING – SPECIALIZATION IN ENVIRONMENTAL ENGINEERING at *Faculty of Exact, Chemical and Natural Sciences – National University of Misiones*